

3-D Parachute simulation by the immersed boundary method

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ABSTRACT

We apply the immersed boundary (IB) method to the 3-D parachute problem, which involves an interaction between the flexible, elastic, parachute canopy and the high-speed airflow (relative to the parachute) through which the parachute falls. In order to take into account the effect of mass of both parachute and payload, the penalty immersed boundary (pIB) method is used. We construct a parachute model with a flexible fiber-reinforced canopy, a vent at the top, and a spherical payload, and we use the pIB method to study its initial inflation, terminal descent, and soft landing, as well as the interaction between two separate parachutes, and also the case of multiple parachute canopies supporting a single payload. The pIB method has the generality to cover all of these scenarios within a unified simulation framework.

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1. Introduction

The purpose of this paper is to show that the immersed boundary (IB) method can be properly applied to the 3-D parachute problem. The authors have already shown that the IB method is applicable to the 2-D parachute problem [15]. Here we extend and generalize the method and idea used in [15] to the three-dimensional parachute problem. In [15], our focus was on the modeling and simulation of canopy porosity, and on the role of canopy porosity in stabilizing the parachute motion. Here, we obtain a similar effect by providing a vent at the top of the parachute so that part of the airflow can go through the parachute canopy instead of around it. Because the vent is explicitly modeled as a hole in the canopy, we do not need to make use of the concept of canopy porosity as developed in [15].

Parachute aerodynamics are complicated by the intense interaction of the parachute with the flow. Unlike most aerodynamic design, the goal of parachute design is not to reduce drag but to increase it, and this in itself practically guarantees that the flow will be very disturbed by the parachute, with intense vortex shedding and a complicated wake. The cloth part of the parachute, which is called the canopy, changes its shape rapidly in response to the surrounding flow field, but the airflow that generates the aerodynamic forces depends on the shape of the parachute canopy. Thus, parachute aerodynamics is inherently a fluid–structure interaction phenomenon, and the incorporation of this effect requires the time-dependent position of the parachute as well as the usual variables that appear in the Navier–Stokes equations [4,19,22,26].

There are two types of numerical tools commonly used for the parachute problem: conventional grid-based computational fluid

dynamics models [27–31] and vortex element methods [6,32]. Recently, Stein et al. have applied to the 3-D parachute problem a special type of the finite element scheme which is known to be useful for fluid–structure interaction problems. They have combined this scheme with a structural dynamics model and mesh updating method, and they have successively simulated several types of 3-D parachute in the characteristic stages of the parachute dynamics: initial inflation, terminal descent, soft landing, and interaction between two parachutes [29–31].

Here we use a different scheme (the IB method) to simulate parachute dynamics in similar circumstances to those in [29–31]. The IB method was developed to study flow patterns around heart valves, and is a generally useful method for problems in which elastic materials interact with a viscous incompressible fluid. In the IB formulation, the action of the elastic canopy immersed in the air flow appears as a localized body force acting on the fluid. This body force arises from the elastic stresses in the parachute canopy. Moreover, the parachute canopy is required to move at the local fluid velocity as a consequence of the no-slip condition. The central idea of the IB method is that the Navier–Stokes solver does not need to know anything about the complicated time-dependent geometry of the elastic boundary, since that immersed boundary is represented in the fluid dynamics computation as a field of force, and that therefore we can escape from the difficulties caused by the interaction between the elastic boundary and the fluid flow. This whole approach has been applied successfully to problems of blood flow in the heart [20,23–25,33], wave propagation in the cochlea [2,11], platelet aggregation during blood clotting [9], and several other problems [1,8,10,13–15,21].

In order to explore the dynamics of a falling parachute, we need to take into account the mass of the parachute, especially of the payload. In most previous applications of the IB method, however, it has been assumed that the immersed elastic materials are either

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massless (in the case of surfaces) or neutrally buoyant (in the case of immersed structures that occupy nonzero volume). This is a good enough approximation in many cases. In some problems, however, the mass of the immersed material can be crucial and affect the overall dynamics [16,34]. In order to provide mass to the canopy and suspension lines as well as to the payload, we use the penalty immersed boundary (pIB) method, which was designed by the authors and applied in [16] to several problems in which the mass of the boundary is important. An empirical demonstration of the convergence of the pIB method is also included in [16]. See also [7,13,17] for more examples of the application of the pIB method.

The basic idea of the pIB method is to add mass to an otherwise massless boundary by connecting massive particles to the massless boundary with stiff springs. As in all IB method computations, the massless boundary moves at the local fluid velocity and applies forces locally to the fluid. These forces are generated by the elasticity of the massless immersed boundary and, in the case of the pIB method, by the springs that link the massive particles to the massless boundary. Meanwhile, at the other end of these same springs, the massive particles move as though in a vacuum according to Newton's law of motion, in which the only forces that appear are the force of gravity acting on each massive particle and the force of the spring that connects that massive particle to the massless immersed boundary. The penalty parameter of the pIB method is the stiffness that is assigned to the springs that link the massless boundary to the added massive particles. As this stiffness tends to infinity, the massive particles track the motion of the massless boundary and endow it with mass. This behavior has been verified quantitatively in [16]. Note that the pIB method is easy to implement in the context of an existing IB code for the massless (neutrally buoyant) case. In particular, the same constant-density Navier–Stokes solver can be used, since the added mass is not attributed to the fluid, as in [34].

An important limitation of the present work is that the Reynolds number of the computations reported here are in the hundreds, whereas real parachutes have Reynolds numbers in the hundreds of thousands, or even higher. We emphasize that this limitation is not a fundamental one, but results from the grid resolution that is practical to achieve on present-day computers. The same method that is proposed here can be applied at higher Reynolds number on a finer grid, and there is no fundamental limitation on the Reynolds number that can be achieved in this way. In practice, however, straightforward mesh refinement is not the most likely route to bridging the three to four order of magnitude gap in Reynolds number described above. More promising strategies include adaptive mesh refinement and turbulence modeling, both of which are being actively pursued within the context of the immersed boundary method [5,12]. Therefore, even though the specific results of this paper cannot be taken seriously for practical parachute design, we are optimistic that the methods presented here can and will be used for that purpose in much the same manner as in the present paper, but in the context of an immersed boundary method for high Reynolds number flow based on some combination of the strategies mentioned above.

The rest of the paper is organized as follows. We describe the equations of motion of the pIB method in Sections 2 and 3 and its numerical implementation including external boundary conditions in Section 4. In Section 5, we construct and specify the 3-D parachute model and present the initial configuration, physical and numerical parameters. Section 6 shows the simulation results which include the inflation of a parachute, the landing of a parachute on the ground, comparison of drag coefficients of the payload in three different situations, and the interactions of two side-by-side parachutes. The conclusions follow in Section 7.

2. Equations of motion

We begin by stating the mathematical formulation of the equations of the motion for a system comprised of a three-dimensional viscous incompressible fluid containing an immersed, elastic, massive boundary.

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

$$\mathbf{F} = -\frac{\partial E}{\partial \mathbf{X}} + \mathbf{F}_M, \quad (3)$$

$$\mathbf{F}_M = -M \frac{\partial^2 \mathbf{X}}{\partial t^2} - M g \mathbf{e}_3, \quad (4)$$

$$\mathbf{f}(\mathbf{x}, t) = \int \mathbf{F}(r, s, t) \delta(\mathbf{x} - \mathbf{X}(r, s, t)) dr ds, \quad (5)$$

$$\frac{\partial \mathbf{X}}{\partial t}(r, s, t) = \mathbf{u}(\mathbf{X}(r, s, t), t) = \int \mathbf{u}(\mathbf{x}, t) \delta(\mathbf{x} - \mathbf{X}(r, s, t)) d\mathbf{x}. \quad (6)$$

Eqs. (1) and (2) are the familiar Navier–Stokes equations for a viscous incompressible fluid. The constant parameters ρ and μ are the fluid density and viscosity, respectively. The unknown functions in the fluid equations are the fluid velocity, $\mathbf{u}(\mathbf{x}, t)$; the fluid pressure, $p(\mathbf{x}, t)$; and the force per unit volume applied by the immersed boundary to the fluid, $\mathbf{f}(\mathbf{x}, t)$; where $\mathbf{x} = (x, y, z)$ are fixed Cartesian coordinates, and t is the time.

Eqs. (3) and (4) are the immersed boundary equations which are written in Lagrangian form. The unknown $\mathbf{X}(r, s, t)$ completely describes the motion of the immersed boundary, and also its spatial configuration at any given time t . Note that $\mathbf{X}(r, s, t)$ represents a 2-D surface in 3-D space, which is used for both the canopy and payload here. Suspension lines and risers are modeled in a similar way, but as space curves, so one of the two parameters r, s is deleted in the case of these parachute components. In Eq. (3), $\mathbf{F} = \mathbf{F}(r, s, t)$ is the force density applied by the immersed boundary to the fluid, in the sense that $\mathbf{F}(r, s, t) dr ds$ is the force applied to the fluid by the patch of immersed boundary $dr ds$. The elastic contribution to this force density is given by the variational derivative $-\frac{\partial E}{\partial \mathbf{X}}$ of the elastic energy functional $E[\mathbf{X}(\cdot, \cdot, t)]$. This variational derivative is implicitly defined by

$$dE(t) = \int \frac{\partial E}{\partial \mathbf{X}}(r, s, t) \cdot d\mathbf{X}(r, s, t) dr ds, \quad (7)$$

where $d\mathbf{X}$ is a perturbation of the boundary configuration and dE is the resulting perturbation in the elastic energy of the boundary (to first order).

In this paper, the energy functional $E[\mathbf{X}(\cdot, \cdot, t)]$ is chosen in a simple way. The 2-D immersed boundaries are modeled as collections of 1-D strings each of which resists stretching and compression:

$$E[\mathbf{X}(\cdot, \cdot, t)] = \frac{1}{2} c_s \int \left(\left| \frac{\partial \mathbf{X}}{\partial s} \right| - 1 \right)^2 dr ds + \frac{1}{2} c_r \int \left(\left| \frac{\partial \mathbf{X}}{\partial r} \right| - 1 \right)^2 dr ds \quad (8)$$

where c_s and c_r are constants. Since, for a fixed r or s , $\mathbf{X}(r, s)$ is a string which is parameterized as a function of s or r , respectively, each such curve can be assigned the same strain–stress relation as in Eq. (8). Finally all the forces generated from all these strings are summed up to make the elastic force.

The contribution of the boundary mass to the force density applied by boundary to the fluid in which it is immersed is denoted $\mathbf{F}_M$ in Eq. (3) and is written out in Eq. (4), where $M = M(r, s)$ is the mass density of the immersed boundary in the sense that $M(r, s) dr ds$ is the mass of the patch of boundary $dr ds$. The two terms on the right hand side of Eq. (4) are, of course, the inertial force (sometimes called the D'Alembert force) and the gravitational force. In the latter, $\mathbf{e}_3$ is the unit vector pointing “up”, i.e. against

gravity. Note that Eqs. (3) and (4) can be combined to yield Newton's law of motion for the immersed boundary:

$$M \frac{\partial^2 \mathbf{X}}{\partial t^2} = -\frac{\partial E}{\partial \mathbf{X}} - \mathbf{F} - M g \mathbf{e}_3. \quad (9)$$

Since $\mathbf{F} d\mathbf{r}_{ds}$ is (by definition) the force applied by the immersed boundary to the fluid, $-\mathbf{F} d\mathbf{r}_{ds}$ is (according to Newton) the force applied by the fluid to the immersed boundary. Although $-\mathbf{F}$ could be expressed in terms of the jump in the fluid stress tensor across the boundary, we shall have no need to do so.

Eqs. (5) and (6) which are called interaction equations involve the three-dimensional Dirac delta function $\delta(\mathbf{x}) = \delta(x)\delta(y)\delta(z)$, which expresses the local character of the interaction. Eq. (5) expresses the relation between the two corresponding force densities $\mathbf{f}(\mathbf{x}, t) d\mathbf{x}$ and $\mathbf{F}(r, s, t) d\mathbf{r}_{ds}$. We can see this fact by integrating each side of the Eq. (5) over an arbitrary region Ω . Note, however, that $\mathbf{f}(\mathbf{x}, t)$ is singular like a one-dimensional delta function with the singularity supported on the immersed boundary. This is because the delta function that we use is the three-dimensional but the integral in Eq. (5) is only two-dimensional. Thus $\mathbf{f}(\mathbf{x}, t)$ is infinite on the immersed boundary and zero elsewhere, but in such a manner that its integral over finite volumes is finite.

Eq. (6) is the equation of motion of the immersed elastic boundary. It is just the no-slip condition which says that the boundary moves at the local fluid velocity. This is rewritten in terms of the Dirac delta function in the second form of Eq. (6). We do so in order to expose a certain symmetry with Eq. (5), in which the force generated by the immersed boundary is re-expressed as body force acting on the fluid. This symmetry is important in the construction of our numerical scheme. Note, however, that the integral in Eq. (6) is a triple integral ($dx_1 dx_2 dx_3$), unlike the integral in Eq. (5). Thus $\partial \mathbf{X} / \partial t$ is finite, unlike $\mathbf{f}$.

3. Mathematical formulation of the penalty immersed boundary method

Now we modify Eqs. (1)–(6) to make the penalty IB method, with which we can handle the mass of the elastic material efficiently. The method described in this section is used to give mass to all of the parachute components, including the canopy, suspension lines, risers, and payload. In the pIB method, we use a dual representation of the immersed boundary. One of its representatives, denoted $\mathbf{X}(r, s, t)$, is massless and elastic and moves at the local fluid velocity, just as in the case of an immersed boundary in a more traditional IB method computation. The other representative, denoted $\mathbf{Y}(r, s, t)$, carries all the mass and is linked to $\mathbf{X}(r, s, t)$ by a system of stiff springs. The different points of the massive boundary are not coupled directly to each other or to the fluid. They move

as though in a vacuum according to Newton's law of motion, with the only forces acting on such mass points being those of gravity, and those arising in the springs that link the two boundaries, see Fig. 1.

In mathematical terms, we get the pIB method by replacing Eqs. (3)–(4) by the following:

$$\mathbf{F} = -\partial E / \partial \mathbf{X} + \mathbf{F}_K, \quad (10)$$

$$\mathbf{F}_K(r, s, t) = K(\mathbf{Y}(r, s, t) - \mathbf{X}(r, s, t)), \quad (11)$$

$$M(r, s) \frac{\partial^2 \mathbf{Y}(r, s, t)}{\partial t^2} = -\mathbf{F}_K(r, s, t) - M(r, s) g \mathbf{e}_3. \quad (12)$$

The first of these equations defines the force density $\mathbf{F}$ which is transmitted by the massless boundary to the fluid. It includes the force density $\mathbf{F}_K$ that is defined by the second of the above equations and is the force density generated by the stiff springs that link the massive and massless representations of the immersed boundary. The stiffness parameter K is the “penalty” parameter of the method. The larger it is, the greater the energy penalty that must be paid to separate the massive and massless boundaries by any given amount. The limit $K \rightarrow \infty$ will be discussed below. Finally the third of the above equations is the equation of motion of the massive boundary, which carries the mass density $M(r, s)$ and has its motion described by the function $\mathbf{Y}(r, s, t)$. Note that the only force densities in the equation of motion of the massive boundary are the penalty force density and gravitational force density. The massive boundary does not interact with the fluid directly.

Consider the case in which K in Eq. (11) goes to infinity. Then the massless boundary $\mathbf{X}(r, s, t)$ and the massive boundary $\mathbf{Y}(r, s, t)$ coincide, and $\mathbf{F}_K$ in Eq. (12) approaches $\mathbf{F}_M$ in Eq. (4). In practice K cannot be infinite but we can keep $\mathbf{X}(r, s, t)$ and $\mathbf{Y}(r, s, t)$ as close as we like by choosing K sufficiently large.

In practice there is a stability restriction on K that depends on the time step duration Δt . It is empirically of the form $K \Delta t^2 < \text{constant}$. Thus, in a convergence study, the parameter K can be allowed to approach infinity as Δt approaches zero. In fact, the situation is quite favorable in that K can be multiplied by 4 every time when Δt is reduced by a factor of 2. For an example of such a convergence study, see [16].

In order to introduce a massive payload, which is a ball in our simulation, we modify pIB method explained above. We assume that the mass of the ball is concentrated at its center (point mass). Then the massless boundary $\mathbf{X}(r, s, t)$ is a sphere (boundary of the ball) and the massive boundary $\mathbf{Y}(t)$ is a point (center of the ball). When we previously assumed that the two boundaries were supposed to be the same, we used $\mathbf{F}_K(r, s, t) = K(\mathbf{Y}(r, s, t) - \mathbf{X}(r, s, t))$ (Eq. (11)), which acts to keep the distance between two boundaries as small as possible. Here, we want to maintain a distance R , the

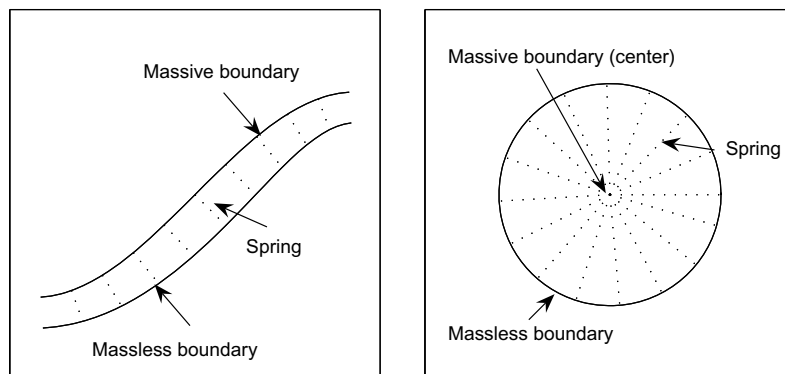


Fig. 1. Massive and massless boundaries are linked together by a system of very stiff springs of which the rest length is zero (left) or the radius of the parachute payload, which is shown in cross-section (right).

radius of the spherical payload, between the surface of the payload and its center. Therefore, we replace Eq. (11) by the following, in which R appears as the rest length of the penalty springs:

$$\mathbf{F}_K(r, s, t) = K(|\mathbf{Y}(t) - \mathbf{X}(r, s, t)| - R)(\mathbf{Y}(t) - \mathbf{X}(r, s, t)). \quad (13)$$

Here KR is the spring constant, and the direction of the force $\mathbf{F}_K(r, s, t)$ is from $\mathbf{X}(r, s, t)$ to $\mathbf{Y}(t)$. See the right panel of Fig. 1 which shows a 2-D simplification of the ball in 3-D.

For the motion of the center, we need to sum all the reaction forces from the spherical massless boundary onto the center, i.e.,

$$M \frac{\partial^2 \mathbf{Y}(t)}{\partial t^2} = - \int \mathbf{F}_K(r, s, t) dr ds - M g \mathbf{e}_3, \quad (14)$$

where M is the point mass and the integration is done over the sphere. When we deal with the force and motion related to the payload, we use Eqs. (13) and (14) instead of Eqs. (11) and (12), which are used for the force and motion of the massive canopy and the suspension lines (except that the suspension lines are only one-dimensional, so one of the two parameters r, s has to be deleted when Eqs. (11) and (12) are applied to them).

We have seen so far three different methods to handle mass of an elastic boundary in the context of the IB method. These are the mass-spreading method [34], the direct use of the D'Alembert force [18], and the pIB method of the present paper. A preliminary comparison is presented in [17,18] where, for a particular test problem, the simulation results by the three methods are qualitatively comparable. We can also find two verifications for the pIB method in [16]. One is a numerical convergence study which shows that the pIB method is 2nd order accurate for a particular test problem. The other is direct comparison with experimental data of vortex-induced vibrations of a massive cylinder, which shows that the results obtained by the present method are quite comparable to the experimental data.

4. Numerical implementation and external boundary conditions

What has been stated so far is the mathematical formulation of the pIB method. For its numerical implementation, we use a 'formally' second-order IB method which is described in [25], generalized to take into account the massive boundary that is linked to the immersed elastic boundary by stiff springs [16]. In this method, each time step proceeds in two substeps, which are called the preliminary and final substeps. In the preliminary substep, we get data at time level $n + \frac{1}{2}$ from data at time level n by a first-order accurate method. Then the final substep starts again at time level n and proceeds to time level $n + 1$ by a second-order accurate method. This Runge-Kutta framework allows the second-order accuracy of the final substep to be the overall accuracy of the scheme. The step-by-step procedure of the numerical implementation can be summarized as the followings:

- (1) Update the position of the massless boundary at time level $n + \frac{1}{2}$. This is done by the discretization of Eq. (6). At the same time, the position of the massive boundary should evolve in the same fashion according to the velocity of the massive boundary.
- (2) Using these new positions, calculate the Lagrangian force density which is the sum of two parts: one is elastic force and the other is from the spring linked between massless and massive boundaries, see Eqs. (10) and (11).
- (3) Change this elastic force density defined on Lagrangian grid points into the force at Eulerian spatial grid points to be applied in the Navier–Stokes equations. This is done by a discretization of Eq. (5).

- (4) Given the Eulerian force density, we are ready to solve the discretized version of the fluid Eqs. (1) and (2) at time level $n + \frac{1}{2}$. The velocity of the massive boundary is also calculated in the same fashion, see Eq. (12).
- (5) Update the positions of massless and massive boundaries at time level $n + 1$ in the same manner as in (1).
- (6) Update the fluid velocity data by solving the fluid equation and the velocity of the massive boundary at time level $n + 1$. This completes the time step.

In solving the fluid equations in both (4) and (6), we use the discrete Fourier transform (implemented by the FFT algorithm) together with the assumption of periodic boundary conditions. Note, however, that there is no fundamental requirement of using a periodic domain in conjunction with the pIB method. Any uniform grid fluid solver based on finite element or finite difference methods can be used, with whatever boundary conditions that solver can accommodate. We have emphasized the periodic case because of the efficiency that follows from the use of the FFT algorithm for solving the linear systems that arise in our numerical scheme.

To complete the description of the numerical pIB method, we need to explain the boundary conditions imposed along the edges of our computational domain. Both for computational convenience, and also to avoid wall effects, we use periodic boundary conditions in all three space directions, but we “break” the vertical periodicity in two different ways depending on the circumstances that we want to simulate.

In most of the simulations of this paper, we generate an updraft to keep the parachute within the computational domain. The updraft is generated within a rectangular slab that spans the bottom of the domain by a body force which is applied there and is adjusted locally to achieve a desired velocity within the slab. The desired velocity, in turn, is adjusted by a control mechanism that senses the height of the parachute payload and increases or decreases the updraft accordingly. The equation governing the control mechanism is as follows:

$$\frac{dW_0(t)}{dt} = \gamma(Z_{\text{target}} - Z_p(t)) - \sigma W_p(t). \quad (15)$$

Here, $Z_p(t)$ and $W_p(t)$ are obtained by taking the z -components of the position and velocity of the center of the payload, respectively. These values can be calculated by Eq. (14) in each time iteration. The velocity $(0, 0, W_0(t))$ is the updraft at time t , Z_{target} is the fixed value at which the z -coordinate of the payload center is supposed to be in its equilibrium state, and γ and σ are constant coefficients. Note that the change of the updraft velocity is damped according to $W_p(t)$ in order to avoid large oscillations of the inflow velocity.

Besides keeping the simulated parachute within the computational domain, the control of the velocity within the slab at the bottom of the domain has the effect of erasing the features of the wake of the parachute that otherwise would leave the periodic domain at the top and re-enter it at the bottom. Thus, we provide a nearly uniform updraft from below through which the simulated parachute falls.

The only exception to the above is when we want to simulate the actual landing of a parachute. In that case, we need to provide a simulated ground on which the payload, and then the parachute itself, can land. This is done by putting “target points” in the position where we want the ground to be. To avoid leaks, the target points should be spaced about half a meshwidth apart (or closer). The target points do not move, and they do not interact with the fluid directly, but each of them is connected by a stiff spring to an immersed boundary point that moves at the local fluid velocity and applies force locally to the fluid. The net effect is to keep the

immersed boundary close to the target points, thus modeling the ground. With this methodology, it is easy to model arbitrary topography of the ground on which the parachute lands, since the topography is specified merely by choosing the coordinates of a list of target points that mark the position of the ground. Examples of parachutes landing on sloping ground will be presented below. Note the close analogy between the use of target points to model a fixed boundary and the use of the pIB method to model a massive boundary. The target points here are like the massive particles of the pIB method, with the simplification that the target points do not move.

5. Three-dimensional parachute model

The model of the parachute which we use here is a conical ribbon parachute which is most widely manufactured and used in practice [4,19,22,26].

The basic component of the conical parachute is an isosceles triangle which is called a gore. The several gores are put together and seamed with each other along their sides to make one canopy (cloth part of the parachute), see Fig. 2. One gore is made of three kinds of fabrics or ribbons: one is a vertical line, another a horizontal line, and the other a radial line. These lines are different in their dimensions and strengths. Because a falling parachute in air experiences different drag forces in different parts, the part that needs to exert a larger force needs a stronger line or ribbon. Our computational model can take this into account by changing the stiffness coefficients of these lines.

Many real parachutes have gaps between ribbons to make the canopy porous. In computation, however, it is impractical to resolve these gaps individually. To do so would require that each gap be at least a few fluid meshwidths in diameter. Otherwise the hole would be effectively closed. Thus a very fine mesh would be required. Because of this difficulty, we do not make gaps between the ribbons except for one big hole called a vent at the top of the canopy, see Fig. 3.

Although we do not do so here, note that one can model a porous canopy within the IB or pIB framework by allowing the normal velocity of the canopy to differ from that of the nearby fluid by an amount proportional to the normal component of the boundary force. For details of this formulation of the porous canopy problem, and for results on the role of porosity in stabilizing the parachute motion, see [15].

The vent on the top of the real parachutes is known to be important both to stabilize the parachute motion and to assist in its secure early inflation [26]. Another benefit of the vent is to prevent potentially catastrophic force concentration as the suspension lines converge towards each other at the “north pole” of the parachute.

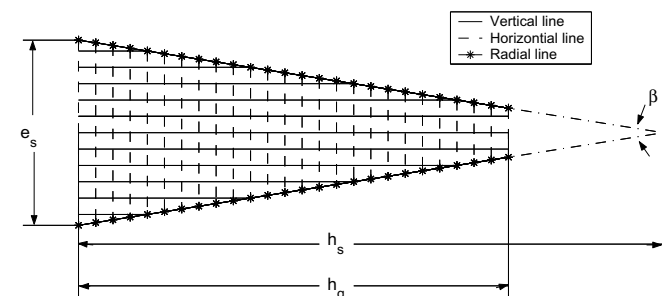


Fig. 2. Gore layout. The three types of lines that comprise one gore of the parachute canopy are shown, together with the parameters that determine the geometry of the gore, and hence that of the parachute canopy, once the number of gores has been chosen.

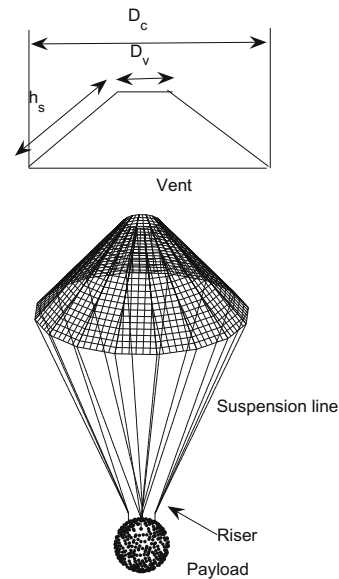


Fig. 3. Constructed profile of a parachute (top) and the complete conical parachute model with canopy and payload (ball) which are connected by suspension lines and risers (bottom).

We attach N_g gores of the same size side by side to make a cone shaped canopy (Fig. 3). For a canopy to have a conical shape, the only criterion needed is $\beta < 2\pi/N_g$, where β is the vertex angle of the gore (Fig. 2 and Table 1). The top picture in Fig. 3 shows the constructed profile of a conical parachute, and the bottom one shows the completed model of the canopy with other elements of the parachute: suspension lines, risers, and payload. One end of each suspension lines is attached to a corner vertex of each gore of the canopy, and the other ends of four consecutive suspension lines are put together and connected to one end of a riser. The risers are now attached to the surface of the payload, see Fig. 3.

The suspension lines and risers play the same dynamical roles in the fluid–structure interaction as those of the canopy and payload, i.e., they have mass and interact along their lines with the surrounding fluid. This is done by the unified idea (pIB method) described in Section 3 and used for all the simulations in this paper. Since, however, the suspension lines and risers are expressed as 1D curves, we need to modify the equations of motion concerning these lines by reducing the dimension of the immersed boundary. In contrast to the suspension lines and risers, the lines that appear to run radially through the vent at the top of the parachute in the figures that follow do not actually interfere with flow in the vent at all, but merely transmit force from one side of the vent to the other.

In Table 1, we specify the geometrical parameters of the model parachute. The terminology used is that of the industry. Figs. 2 and

Table 1
Construction parameters of parachute.

Parameter	Symbol	Magnitude
Height of triangle of each gore	h_s	0.42 m
Height of gore	h_g	0.3757 m
Number of gores	N_g	16
Gore base length	e_s	0.1106 m
Vertex angle of gore	β	$\pi/12$
Base cone angle	μ	0.27π
Length of cone base	D_c	0.2780 m
Length of vent	D_v	0.0443 m
Suspension line length	l_s	0.62 m
Riser length	r_s	0.03 m
Radius of payload	R	0.07 m

3 show how these parameters determine the geometry of the model parachute. The overall size of the model parachute that we simulate is small, but the ratios of vent size and suspension lines to the canopy size are comparable to those of a real parachute.

Table 2 shows the physical parameters used in the simulation. To avoid the consideration of turbulent flow, we reduce the Reynolds number from about 300,000 (which would be typical of a real parachute) to about 300. This is done in part by considering a small parachute, as mentioned above, but also by artificially increasing the viscosity of the air. Strategies for meeting this challenge are discussed in Sections 1 and 7.

In all the simulations in this paper, the coefficients of stiffness in Eq. (8) are $c_r = c_s = 100$ N/m for the canopy and $c_s = 10$ N for the suspension lines and risers. The penalty parameter K is chosen to keep the distance between the massless and massive boundaries as

close as their nominal distance. With $K = 5000$ kg/(m s)² which we use, the maximum deviation of the distance between the two boundaries stays within one hundredth of the meshwidth H . The meshwidth $H = \Delta x = \Delta y = \Delta z = 2.4/128$ is uniform and fixed in time. The final time T is normally 10.0s and the constant timestep is 0.0001 s which requires 100,000 time steps to complete one simulation. A typical computation on a $128 \times 128 \times 128$ fluid grid with about 40,000 nodes representing the parachute, and covering about 10^5 time steps, requires 120 h to complete on a 2.2 GHz opteron processor.

6. Results and discussion

6.1. Inflation and landing of a parachute

The first computational experiment is to study the change of shape of the parachute canopy when it inflates, beginning from a nearly closed configuration and ending at a later time with the fully open canopy, see Fig. 4. Of all the results that we consider, it is this one that shows most clearly the need for a method that can handle the unknown changes in shape of the parachute canopy. In the IB method, this is done without any re-gridding, since the canopy is represented in the fluid dynamics computation by a force field defined on a uniform grid. For alternative approaches, see [6,27].

In Fig. 4, the deformation of the canopy and the fluid motion around it are shown during the inflation of the parachute as the time goes on. The initial configuration is modified from Fig. 3 to be of the shape of a cone with a very small opening, see the left-

Table 2
Physical parameters of parachute simulations.

Physical parameters	Symbol	Magnitude	Unit
Density	ρ	1.2	kg/m ³
Viscosity	μ	2×10^{-3}	kg/(m s)
Gravitational acceleration	g	9.8	m/s ²
Mass of payload	M_p	0.004–0.012	kg
Density of canopy	M_c	0.005	kg/m ²
Density of suspension Line (Riser)	M_s	0.0003	kg/m
Updraft velocity	W_0	0.6–1.2	m/s
Reynolds number		Around 300	
Computational domain		$2.4 \times 2.4 \times 2.4$	m $\times$ m $\times$ m

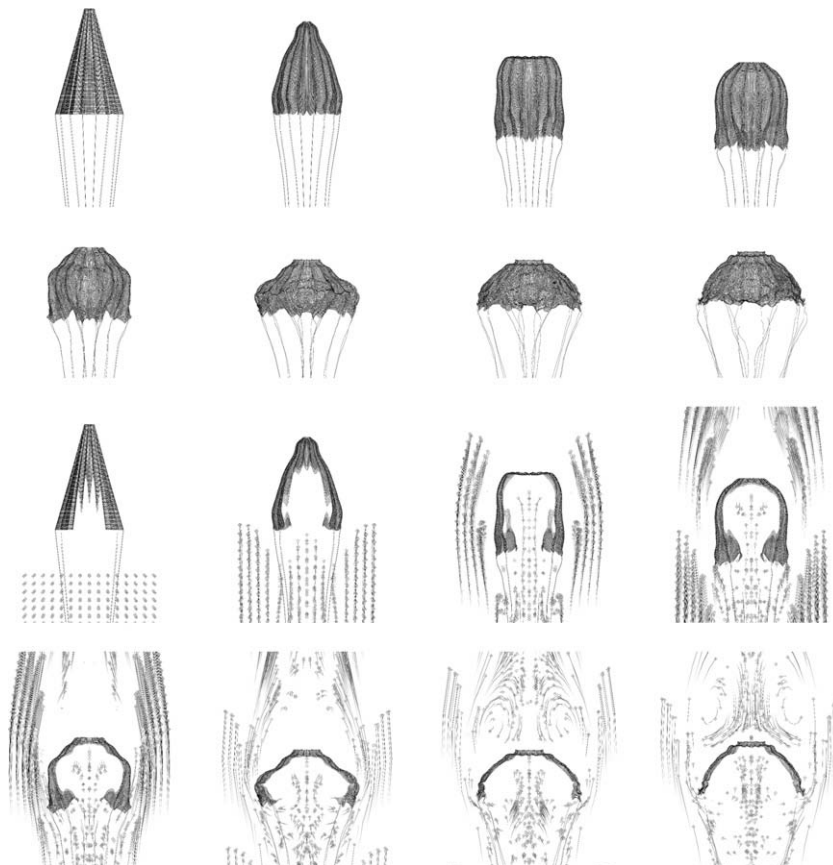


Fig. 4. The inflation of the parachute is shown at a series of times: 0, 0.4, 0.8, and 1.2 s in the first row, and 1.6, 2.0, 2.4, and 2.8 s in the second row. Note the gradual formation of arches along the lower edge of the parachute canopy between the suspension lines as the parachute inflates. The bottom two rows of the figure show the corresponding cross-sections of the parachute at the corresponding times to those in the upper two rows, with the recent trajectories (streak lines) of some fluid markers also shown. Note that the markers are plotted as tiny blobs at the current time, with lines showing their recent past as though they had left a trail.

top picture of the figure. The upper two rows show the inflation process of the parachute. At the beginning of the parachute inflation, the canopy begins to inflate from the bottom through the middle to the top, after which all parts of the canopy are equally inflated, see the third frame in the first row of Fig. 4. Air captured by the canopy finally applies enough pressure to the lower part of the canopy so that the parachute inflates completely. This inflation process is comparable to the results in [6,15,27], even though they were 2-D simulations and/or used different methods.

The bottom two rows of Fig. 4 show vertical sections of the parachute at times corresponding to those shown in the first two rows of the figure. From the trails which are left by fluid markers and which show their recent trajectories (streak lines), we can observe that there is flow separation at the canopy edge, that some vortices are forming above the canopy, and that flow within the canopy is pushing the canopy outward. After the time proceeds beyond the last time shown in Fig. 4, the parachute moves stably without further change of its configuration. We can see the motion of the parachute at the later time $t = 8.0$ s in Fig. 5. The cross-section with streak lines in the right panel of Fig. 5 shows clearly the vortex structure of the wake of the payload and also of the parachute itself. It is interesting how the wake of the payload appears to be trapped and stabilized by the parachute canopy. Also note how the wake of the parachute interacts with the small jet that emerges from the vent at the top of the parachute. There is a stagnation point just above the top of the parachute canopy where this jet meets the downdraft of the vortex wake of the parachute as a whole.

It is interesting to see a parachute landing on a fixed ground [31]. For the initial setting of the simulation, we have saved the data obtained from the previous simulation of the falling parachute when the parachute and fluid velocity reach their steady states. The initial condition for the dropping parachute, for example, is

the state represented in Fig. 5. See also the left-top panel of Fig. 6. The initial velocity field, however, is modified by subtracting from it everywhere the updraft velocity that was being imposed near the bottom of the domain in the calculation of the steadily falling parachute. This change is made because the steady fall of the parachute was computed in an updraft just sufficient to hold the parachute at a constant height. By subtracting the velocity of the updraft, we make the vertical velocity near the parachute be negative and the vertical velocity near the ground be close to zero. This is essentially a change of reference frame from one in which the parachute is at rest to a reference frame in which the ground is at rest. The ground on which the parachute lands is of the shape of hills and made by tethering immersed boundary points to fixed target points with stiff springs, as described in Section 4. Note that any shape of ground can be made by this methodology.

Fig. 6 shows the two types of fixed hills and the motion of the landing parachute as the time goes on. After the payload hits the ground, the payload rolls along the hill, see the fourth and fifth figures in both results in which the points on the payload attached to the suspension lines are asymmetric. Because the suspension lines oppose the rolling motion of the payload, this rolling soon stops and is replaced by a sliding motion of the payload, which drags the canopy along as it slides down the hill. See also Fig. 7, which shows enlarged views of the last pictures in both of the cases shown in Fig. 6. Although these results involve several types of collisions (payload with ground, canopy with ground, and canopy with payload), it is important to note that no special collision detection was required during the computation. Non-penetration of immersed boundaries is an automatic feature of the IB method, and follows (if the time step is sufficiently small) from the continuity of the interpolated velocity field in which the immersed boundaries move. Indeed, when collisions seem to occur in an IB computation, there is always a small gap that remains between the colliding surfaces. The gap size is of the same order of magnitude as the meshwidth of the fluid computation.

6.2. Parachute with one and multiple canopies

In this section, we compare the drag coefficients acting on the payload in three different situations: (1) a payload dropping alone, without a parachute; (2) a payload supported by a one-canopy parachute; and (3) a payload supported by a three-canopy parachute.

When a body is placed in and partially blocks a flow, the body feels a drag force. A dimensionless measure of this effect is the drag coefficient C_D which is defined as follows:

$$C_D = \frac{F_D}{\frac{1}{2} \rho V_0^2 A}, \quad (16)$$

where ρ is the density of the fluid, V_0 is the far-field uniform velocity, and A is the projected area of the body which is πR^2 in our case, where R is the radius of the spherical payload. The variable F_D is the total drag force acting on the payload parallel to the direction of the flow. See [3] for more details. Note that the drag coefficient C_D is dimensionless.

The overall performance of a parachute can be summarized by the relationship between the drag force it generates and the speed at which it is falling (relative to the air at a large distance from the parachute). In physical experiments and also in computational simulations, the usual way to determine the drag force and coefficient of a body is to have the body fixed in a given flow. If the payload is fixed in place and the updraft velocity is arbitrary, then the speed of the parachute (relative to the distant air) is the independent variable, and the drag force could be computed. Indeed, the

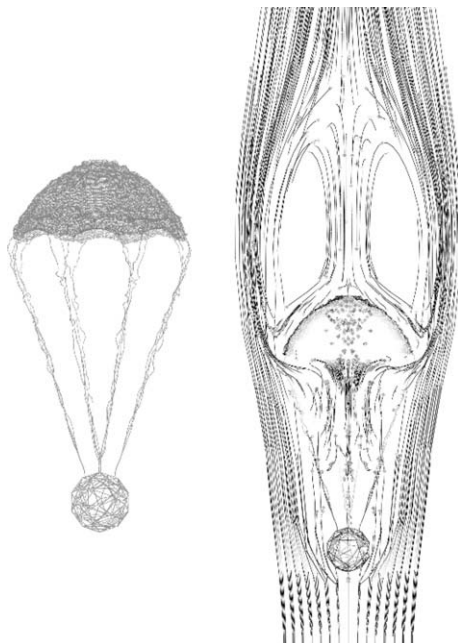


Fig. 5. The fully inflated parachute and its flow pattern at $t = 8.0$ s. The left panel is a perspective view of the parachute and its payload, and the right panel (different scale) is a cross-sectional view with the trails of fluid markers shown to visualize the flow pattern. Note the prominent vortex wake above the payload and again above the parachute canopy. In both cases, there is a strong downdraft on the axis of symmetry (even in our frame of reference, in which the parachute is approximately at rest in an imposed updraft). Also note the updraft through the vent at the top of the parachute canopy.

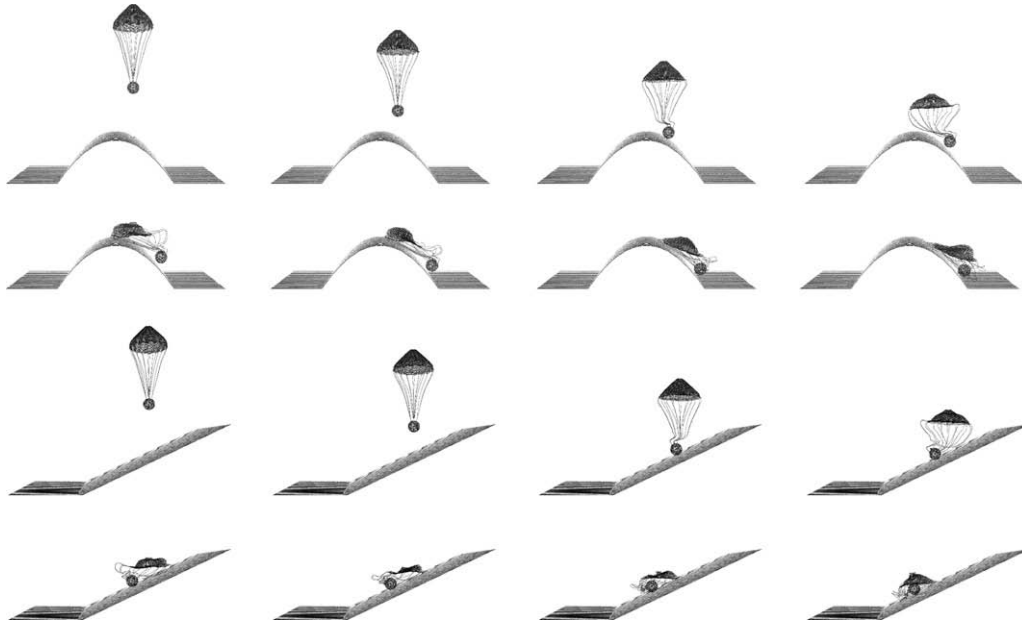


Fig. 6. A parachute is falling by the gravitational force to the ground which is of the shape of hills and made by tethering immersed boundary points to fixed target points with stiff springs. Beginning at the initial setting (left-top), the times are 0, 0.6, 1.2, and 1.8 s in the first row, and 2.4, 3.0, 3.6, and 4.2 s in the second row. The lower two rows show the parachute falling on a different type of hill at the same times in the corresponding positions above.

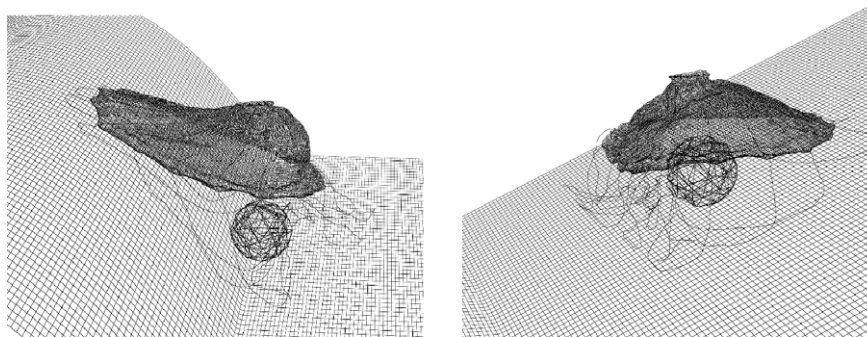


Fig. 7. Zoom-in of the last figures in of Fig. 6. We can observe the collision of the three different immersed boundaries: payload, ground, and canopy.

drag force can be determined simply by examining the tensions and angles of the suspension lines.

In our simulation, however, we drop the payload (in our case, a ball) attached to the parachute and let it fall freely, and the inflow velocity $W_0(t)$ is adjusted to keep the parachute from falling or rising. In this case, the drag force is the independent variable, since it has to be equal to the specified weight of the payload, and the speed corresponding to the given drag force is just the equilibrium value of the inflow velocity, as set by the control mechanism (Eq. (15)). When the position of the payload and the inflow velocity $W_0(t)$ have both settled down, the total force acting on the ball must be zero, and this implies that the total drag force acting on the payload F_D must be equal to the weight of the payload $M_p g$, where M_p is the mass of the payload and g is the gravitational acceleration. The value to which $W_0(t)$ settles down may be thought of as the free stream velocity that generates the drag force F_D .

In order to get better performance, parachute makers often provide parachutes with several canopies supporting a single payload [4,19,22,26]. Adapting the same model of the one-canopy parachute used in the last two sections, we can simply create a three-canopy parachute by connecting one surface point of the payload to three canopies with a riser collecting all the suspension

lines, see Fig. 8. The figure shows the configurations seen from the side (left) and the top (right) at the initial time 0.0 (top) and later time 8.0 s (bottom). We can see that the three canopies are all partially open initially and fully open later. For alternative approaches, see [30] in which the authors have explored the interaction between the multiple canopies.

Before comparing the performances of the parachutes, we first have to check whether our control mechanism (Eq. (15)) results in the desired effect that the controlled updraft settles down to a nearly steady value and succeeds in keeping the center of the payload near the level that we have specified. To see that, we plot the height of the ball center $Z_p(t)$ and the applied inflow velocity $W_0(t)$ in Fig. 9. The dashdot, dashed, and solid lines represent the results of the three cases: (1) the payload only, (2) the payload with one canopy, and (3) the payload with three canopies, respectively. Note that the payload masses are different in the three cases, see below. We can see that the height of the center of the ball in each of the three cases approaches one value $z_{\text{target}} = 0.5$ m (top), and that the inflow velocities stay around values 1.0 m/s in the three cases. This is exactly what the control mechanism is designed to achieve. The parameters $\sigma = 4.0/s^2$ and $\gamma = 5.0/s$ are used for the constants in (15).

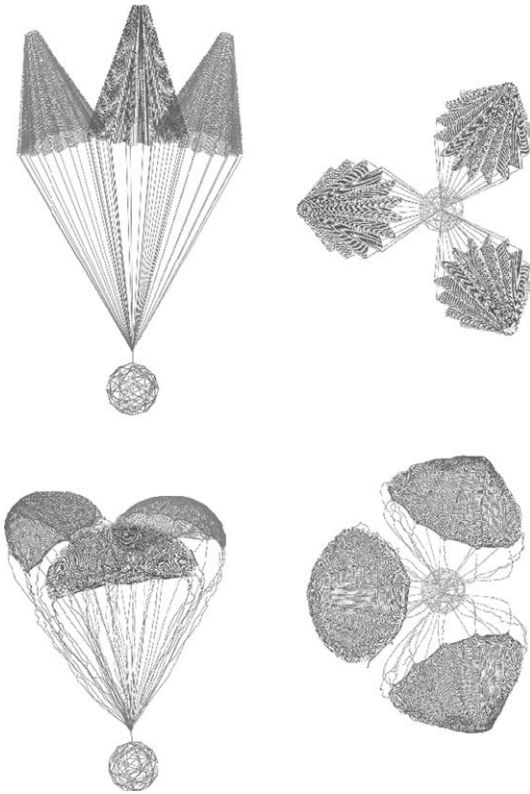


Fig. 8. Three-canopy parachute at times 0.0 s (top) and 8.0 s (bottom).

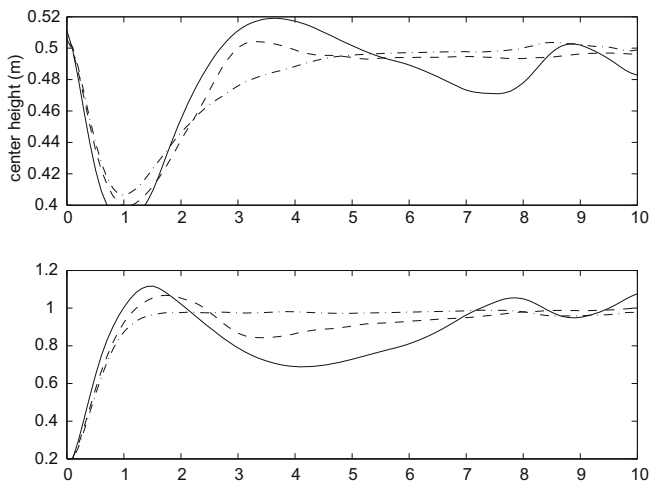


Fig. 9. The height of the ball center $Z_p(t)$ (top) and the desired updraft velocity $W_0(t)$ (bottom) are plotted: dashdot line for the payload only, dashed line for the payload with one canopy, and solid line for the payload with three canopies. The height of the center of the ball in each of the three cases approaches one value $z_{\text{target}} = 0.5$ m (top), and the updraft velocity settles down to values 0.8 m/s in case (1), 1.0 m/s in case (2), and 1.0 m/s in case (3). Note that the payload masses are different in the three cases: 0.004 kg in case (1), 0.006 kg in case (2), and 0.012 kg in case (3).

With the almost uniform velocity $W_0(t)$ shown in the middle panel of Fig. 9, we are ready to calculate the Reynolds number and the drag coefficient. To do that, we also need some additional parameters used in the simulation which were presented in Table 2. To get a reasonable inflow velocity, we use $M_p = 0.004$ kg for the case (1), $M_p = 0.006$ kg for the case (2), and $M_p = 0.012$ kg for the case (3). After the heights of the payloads approach almost steady states, we plot the Reynolds numbers (top) and the drag coefficient

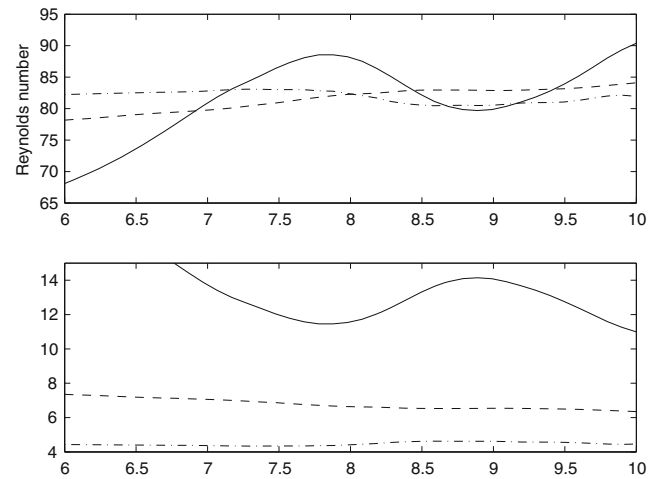


Fig. 10. The Reynolds number (top) and the drag coefficient (bottom) as functions of time in the three cases: payload only (dashdot line), payload with one canopy (dashed line), and payload with three canopies (solid line).

coefficients (bottom) as functions of time in Fig. 10. Note that, while the Reynolds numbers here are based on the diameter of the payload, the actual Reynolds numbers in the simulations are bigger than these values when based on the dimension of the parachute canopy.

The figure shows that the Reynolds numbers are about 80 in all the three cases. The drag coefficients, however, are different: the first case has a drag coefficient of about 4.5, the second one has a drag coefficient of approximately 6.5, and the third one has a drag coefficient of around 12.0.

6.3. Interactions between two parachutes

Finally, we consider the interactions of two parachutes that are falling near each other. When two parachutes get close enough, the motion of one parachute can be strongly influenced by its interaction with the other parachute. These interactions sometimes can lead to the failure of the parachutes such as a chaotically unstable motion and an incomplete inflation. One reason for this failure is that vortex shedding from the lower parachute complicates the flow pattern around the upper one [26,30].

In order to see how the initial distance between two parachutes affects their overall dynamics, two parachutes are initially placed with various displacements in the x -direction from 0.4 to 1.4 m, as measured from the center of one payload to the center of the other payload. In all cases, the initial vertical (z) displacement is 0.8 m, and the initial displacement in the y -direction is zero. After the initial time, both of the parachutes move freely in an updraft velocity that is controlled in the same manner as has been described above by the vertical position and velocity of the payload of the parachute that is initially the lower one. The position and velocity of the other parachute have no effect on the updraft velocity, so the vertical position of that parachute is unconstrained in these simulation. The size of the domain is $3 \times 3 \times 3$ m in these two-parachute simulations.

Fig. 11 shows the differences in the x and y coordinates of the centers of the vents. Let d_x be the displacement in x and d_y be the displacement in y between the centers of the vents of the two parachutes. Each curve represents a different initial x -displacement: 0.4, 0.6, 0.8, 1.0, 1.2, and 1.4 m. As time goes on, both d_x (upper panel) and d_y (lower panel) change dramatically, but whether they increase or decrease is not easy to predict. At early times, there seems to be a threshold distance of about 0.7 m above

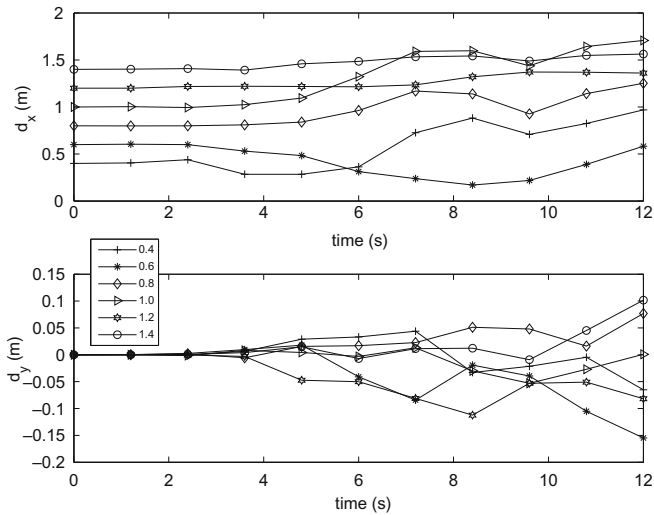


Fig. 11. Displacements d_x in x (upper panel) and d_y (lower panel) in y between the centers of the vents of the two parachutes as functions of time. Each curve represents a different initial d_x . Note the non-monotonic relationship between the final value of d_x and its initial value, and also symmetry breaking that occurs in d_y .

which the distance between the parachutes increases and below which it decreases. In the cases where the parachutes are closest, however, the decrease in the distance between them is only temporary, and the parachutes almost seem to bounce off each other and then to separate quite widely. We can also observe from the

plots of d_y in the lower panel that there is symmetry breaking in the y -direction in all the cases.

Fig. 12 shows the motions of the two parachutes for which the initial displacement in the x -direction is 0.6 m. The upper three rows show the motion of the parachutes seen from the far distance and the lower three rows show the parachutes as seen from above. As time goes on, the parachutes get close, almost collide, and take apart again. This happens only by the large movement of the higher parachute with the lower parachute almost staying at its initial position. When the initial displacement between the two parachutes is slightly bigger than the case shown in Fig. 12, the two parachutes do not seem to interact, and the higher parachute goes almost straight down with keeping the distance from the lower parachute. The critical value of these different behaviors is the initial displacement in the x -direction $d_x = 0.7$ m in our simulation.

7. Summary and conclusions

In this paper, we have shown that the immersed boundary (IB) method is applicable to the study of fluid–structure interaction in parachute aerodynamics. In particular, we have used the penalty immersed boundary (pIB) method, which makes it easy to include both the inertial and gravitational effects of boundary mass in an immersed boundary computation. Here the pIB method is used to give mass to the parachute canopy, suspension lines, risers, and payload.

Some distinctive advantages of the IB methodology are highlighted by these computations. These include its ability to simulate the detailed structure of an immersed boundary (i.e., the particular

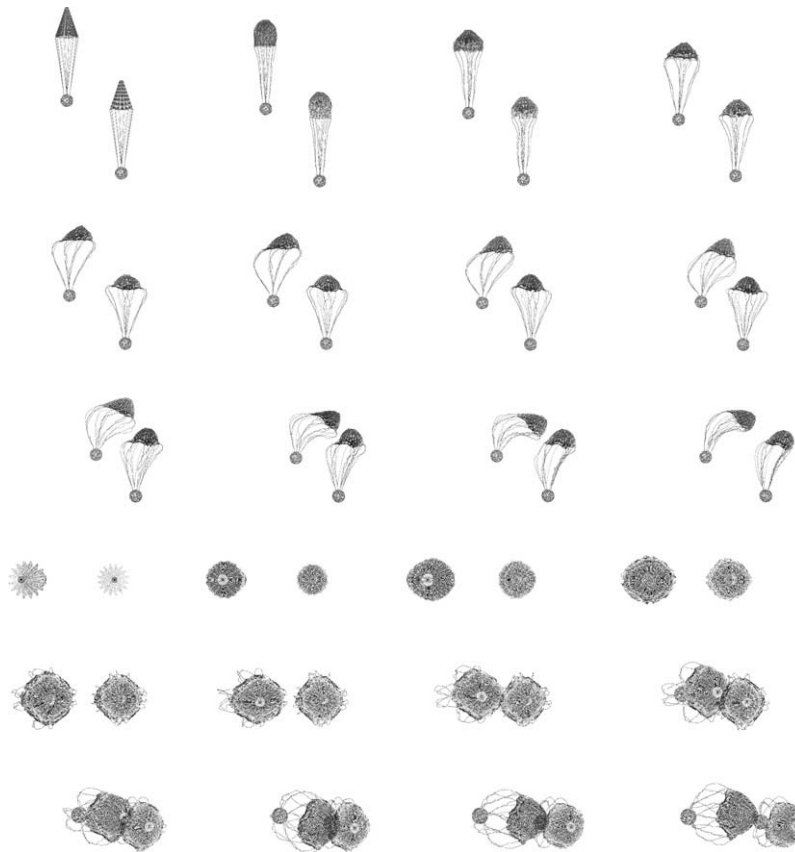


Fig. 12. Interaction of two identical parachutes and payloads. The initial displacement between the centers of their payloads (or vents) is (0.6, 0.0, and 0.8 m). The upper three rows of the figure show the parachutes as seen from the front (i.e., looking along the y -axis); and the lower three rows of the figure show the parachutes as viewed from the top. In each of the two views, the times shown are 0.0, 0.96, 1.92, and 2.88 s in the first row, 3.84, 4.8, 5.76, and 6.72 s in the second row, and 7.68, 8.64, 9.60, and 10.56 s in the third row.

fiber architecture of the parachute); its ability to handle large deformations without re-gridding (as during parachute inflation); and its ability to handle collisions in a physical manner without the use of any special collision detection algorithm (as during parachute landing).

We have shown how the methodology of this paper can be used to study such important problems in parachute aerodynamics as the use of multiple parachute canopies to support a single payload and the interaction of nearby parachutes. In the former case we have shown that the drag per canopy increases with the number of canopies, which illustrates the benefit of attaching several canopies to a single large payload instead of separating the payload into separate parts and letting each be carried by its own canopy. In the latter case, we have demonstrated a symmetry breaking mechanism that helps to avoid collisions between nearby parachutes.

Although the methodology presented here has obvious potential for parachute design by computer simulation, its practical use for that purpose is limited by the low Reynolds numbers (in the hundreds, instead of hundreds of thousands) of the simulations that we have presented. Increases in the Reynolds number may be enabled in various ways, such as improved fluid solvers, global mesh refinement via massively parallel computation, adaptive mesh refinement, direct numerical simulation of turbulence, or the use of turbulence models. It is quite possible, though, that the basic method of parachute simulation that we have described in this paper will still be useful in the context of these more sophisticated approaches to high-speed aerodynamics.

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